

Mohammed Magdy Ismael

Staff Frontend Engineer

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PROFESSIONAL SUMMARY

Staff Frontend Engineer with 8+ years of experience building high-performance, scalable web applications across healthcare, fintech, and e-commerce. Currently at Vezeeta — the Middle East's leading healthcare platform — where I lead frontend architecture, drive AI and automation initiatives, and own measurable product impact at scale. Proven track record integrating LLMs (OpenAI GPT-4, Assistant API, Embeddings), building automation platforms (N8N, ChromaDB, WhatsApp bots), leading company-wide Vite migrations across 5+ projects, and engineering SEO infrastructure across 10+ structured data schema types. Passionate about engineering culture, technical mentorship, and transforming complex challenges into exceptional user experiences.

WORK EXPERIENCE

Staff Software Engineer | Vezeeta

Apr 2026 – Present

Cairo, Egypt | Hybrid | Full-time

- Define and enforce frontend architecture patterns, engineering standards, and technical best practices across all product squads
- Lead performance engineering initiatives — Core Web Vitals, bundle optimization, rendering strategies — across the entire Vezeeta platform
- Drive the AI integration roadmap: LLM-powered workflows, AI agents, and automation pipelines using N8N, ChromaDB, and OpenAI
- Partner with Engineering Management and Product on technical roadmap, cross-team alignment, and architecture decision records
- Run engineering guilds, architecture reviews, and mentorship programs for senior and mid-level engineers
- Champion developer experience: build tooling improvements, monorepo structuring, CI/CD optimization, and infrastructure modernization

Senior Frontend Engineer L2 | Vezeeta

Feb 2024 – Apr 2026

Cairo, Egypt | Hybrid | Full-time

Key Projects:

- **Payment Portal Rewrite:** Redesigned and implemented a centralized payment solution consolidating all payment methods (PayFort, Geidea) across the platform
- **Epharmacy Web Full Rewrite:** Led major platform modernization — Node 10→22, React 16→19, Next.js 9→16 — fully rebuilding the pharmacy e-commerce web application
- **AI Booking Chatbot:** Developed and deployed an AI-powered patient booking chat with real-time patient cards, reviews summary, localization, and operations scripting
- **WhatsApp Bot Ecosystem:** Architected a multi-channel WhatsApp automation system: patient inbound bot, Shamel subscription bot, pharmacy order complaints flow, and collection campaigns via Wassenger + WABA integration
- **N8N Automation Platform:** Designed and deployed 15+ production N8N automation workflows: pharmacy stock updates, MoEngage data sync, ChromaDB ingestion pipelines, Adjust DeepLink creation, outbound/inbound campaigns, and Telegram reporting bots
- **ChromaDB Knowledge Base:** Deployed and managed a ChromaDB vector database for AI-powered semantic search; ingested pharmacy masterlist (170,000 items in 33 min vs. 5 hours previously) and labs/scans masterlist
- **Pharmacy AI Onboarding:** Built a Custom GPT for pharmacy onboarding achieving 93% accuracy; automated the full mapping pipeline — onboarded 50+ pharmacies, processing 8,000 items in under 3 minutes
- **SEO Schema Infrastructure:** Implemented 10+ structured data schema types (Doctor Profile, FAQ, Search Page, Q&A, Reviews, Medical Topics, Local Business, Articles rich results); managed sitemaps, robots.txt, LLMs.txt, and canonical strategy across all domains
- **Twilio VoIP Integration:** Implemented real-time Call over Twilio (VoIP), WebSocket backend, national regulations compliance, and cluster migration

- **Salesforce Integration:** Built doctor-facing Salesforce chatbot scripts; integrated Wassenger with Salesforce for collection notifications and pharmacy order case management
- **OpenAI GPT / Assistant API:** Integrated GPT-4, Assistant API, and OCR for call center automation, explicit content filtering, and ePrescription itemization
- **Pharmacy Marketplace & PMS:** Built pharmacy stock and order management platform; led development of Practice Management System for doctors, clinics, financial reporting, and scheduling

Key Achievements:

- **Core Web Vitals:** INP good URLs: 47.65% → 64.25% (Bad: 31.55% → 13.6%) | LCP good URLs: 74.87% → 83.22% (Bad: 10.89% → 6.67%) | CLS good URLs: 87.99% → 91.71% (Bad: 7.89% → 4.7%)
- **AWS Automation:** Automated pharmacy stock update pipeline (Lambda, SES, S3) — reduced manual effort from 5 hours to 15 minutes
- **Company-Wide Vite Migration:** Led Vite migration across 5+ projects (VOPs, Practice Management, AIApp Web, Epharmacy Web, Questions Portal) — build times from 20–26 min down to under 4 min per project
- **Azure Pipelines Optimization:** Reduced Azure CI/CD pipeline build times from 20–28 minutes to under 3 minutes across all frontend projects
- **EKS Migration:** Migrated all Vezeeta frontend apps to EKS with container routing, redirection strategy, and zero-downtime deployments
- **Multi-Platform Analytics:** Integrated MoEngage, Adjust (S3 reporting + DeepLink automation), TikTok Pixel, Meta Pixel, and GA4 across 5+ platforms and 7 countries (Egypt, Saudi Arabia, Jordan, Lebanon, Kenya, Nigeria)
- **Cloudflare CDN:** Switched JS asset delivery from AWS CloudFront to Cloudflare CDN; set up Cloudflare 301 redirects, firewall notifier, and quota monitoring
- **Performance & Observability:** Implemented Redis caching for Articles pages; set up LangSmith for LLM pipeline observability; building Playwright + OpenAI AI testing framework
- Mentored junior and senior engineers; led technical standards, code reviews, and cross-squad engineering guilds using Agile (SCRUM/Kanban)

Senior Frontend Engineer L1 | Vezeeta

Feb 2023 – Feb 2024

Cairo, Egypt | Hybrid | Full-time

- Spearheaded development of Vezeeta Practice Management System and Pharmacy System (VPS)
- **OpenAI Embeddings:** Integrated OpenAI Embeddings API for NLP-based pharmacy masterlist onboarding — early foundation of the full AI pipeline
- Implemented feature flags using Google Firebase Remote Config for controlled, environment-specific feature rollouts
- **Voffers Upgrade:** Upgraded Vezeeta Offers project from Node 9 to Node 20 across two phases: dependency cleanup and full version upgrade
- Led EKS migration for Practice Management project and Jenkins CI/CD pipelines; migrated Questions Approval Portal from CRA to Vite
- Collaborated with cross-functional teams on SCRUM/Kanban sprints; maintained security and privacy standards across all applications

Frontend Engineer L2 | Vezeeta

Jan 2022 – Feb 2023

Cairo, Egypt | Hybrid | Full-time

- Contributed to Vezeeta/HIS Integration for Saudi Arabia hospitals, enabling seamless bidirectional patient data exchange
- **ePrescription Itemization:** Developed ePrescription itemization using OCR + OpenAI Assistant API to automate pharmacy prescription digitization
- **Azure DevOps:** Configured Azure DevOps CI/CD pipelines, eliminating manual deployment steps and enabling automated multi-branch deployments
- Integrated Raygun for distributed tracing and root-cause analysis; Firebase Remote Config for centralized configuration management
- Optimized web application performance through code refactoring, UI/UX enhancements, and Test-Driven Development (TDD)

Frontend Engineer L1 | Vezeeta

Jan 2019 – Jan 2022

Cairo, Egypt | Full-time

- Played a foundational role in building Vezeeta's Pharmacy Marketplace and Offers Squad from the ground up
- Implemented RESTful APIs and GraphQL for efficient data fetching and state management across consumer-facing products

- Built responsive, accessible UI components using Styled Components and CSS-in-JS
- Collaborated with product and design teams to deliver high-quality experiences across multiple Vezeeta surfaces

R&D Engineer | DevisionX

Jul 2017 – Dec 2018

Heliopolis, Cairo, Egypt | Full-time

- Designed and developed React web applications to support machine learning systems for AI vision inspection and OCR
- **CNN / Deep Learning:** Built and deployed CNN models to extract National ID information from photos, achieving high accuracy in data detection
- Improved Arabic letter segmentation for OCR systems, significantly enhancing Arabic text recognition accuracy
- Developed a reporting system and data warehousing solution for industrial analytics and client decision-making
- Defined solution architectures and delivered custom image processing algorithms tailored to industrial client needs

TECHNICAL SKILLS

Frontend	React.js, Next.js, TypeScript, JavaScript (ES6+), Redux, React Native, AngularJS, Flutter
Styling & UI	Styled Components, CSS-in-JS, Tailwind CSS, SCSS/Sass, Storybook, Figma, Photoshop, UI/UX Design
AI & LLMs	OpenAI SDK (GPT-4, Assistant API, Embeddings), LangChain, Langflow, ChromaDB, LangSmith, NLP
Automation	N8N, WhatsApp Business API (WABA), Wassenger, Playwright, Telegram Bots, Salesforce Integration
Backend	Node.js, Express.js, Python, C# / ASP.net, Prisma, REST, GraphQL, WebSockets, Twilio (VoIP)
Databases	MongoDB, MySQL, PostgreSQL, Redis, ChromaDB (Vector DB)
Cloud & DevOps	AWS (Lambda, S3, SES), EKS, Docker, Kubernetes, Azure DevOps, Jenkins, ArgoCD, Terraform, Cloudflare
Analytics	MoEngage, GA4 / GTM, Mixpanel, Adjust, TikTok Pixel, Meta Pixel, Raygun, LangSmith
Testing	Jest, React Testing Library, TDD, Playwright, Automated Testing
SEO & Performance	Core Web Vitals (INP / LCP / CLS), Structured Data / Schema Markup, Sitemaps, Robots.txt, LLMs.txt, SSR/CSR
Practices	SOLID, OOP, Design Patterns, Agile / SCRUM / Kanban, Code Review, Monorepo (Lerna), Webpack, Vite

EDUCATION

BSc, Computers & Systems Engineering | Mansoura University

Sep 2009 – Jun 2014

Mansoura, Egypt

Nanodegree, Deep Learning | Udacity | 2018

LANGUAGES

Arabic: Native **English:** Full Professional **French:** Intermediate